

Mathematical Methods for Artificial Intelligence

Linear Algebra, Eigenvalues & Eigenvectors

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$$A\vec{v} = \lambda\vec{v}$$

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Transformation matrix

$$\hat{A} \vec{v} = \lambda \vec{v}$$

Transformation
matrix

$$\hat{A} \vec{v} = \lambda \vec{v}$$

Eigenvector

Transformation

matrix Eigenvalue

$$\hat{A} \vec{v} = \hat{\lambda} \vec{v}$$

Eigenvector

Transformation
matrix

Find

Eigenvalue

$$\overbrace{A} \overrightarrow{v} = \overbrace{\lambda} \overrightarrow{v}$$

Find

Eigenvector

$$A\mathbf{v} = \lambda\mathbf{v}$$

we know that $I\mathbf{v} = \mathbf{v}$

$$A\mathbf{v} = \lambda I\mathbf{v}$$

$$\text{or } (A - \lambda I)\mathbf{v} = 0$$

$$A\mathbf{v} = \lambda\mathbf{v} \quad \text{or} \quad (A - \lambda I)\mathbf{v} = 0$$

Eigenvalue Problem

Matrix-vector multiplication

$$\overbrace{A\vec{v}} = \underbrace{\lambda\vec{v}}$$

Scalar multiplication

$$(A - \lambda I)\vec{v} = \vec{0}$$

Solve for
 λ and $\vec{v}$

Solution of Non Homogeneous Linear System

$A\mathbf{x} = \mathbf{b} \implies \mathbf{x} = A^{-1}\mathbf{b}$
provided A^{-1} exists or

$$\det A \neq 0$$

Solving $(A - \lambda I)\mathbf{v} = 0$ or Homogeneous System

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Solution of Homogeneous Linear System

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- if A^{-1} exists or $\det A \neq 0$
- For a **non-trivial** solution

$$\det A = 0$$

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$$\det A = 0$$

For an eigenvalue problem $(A - \lambda I)\mathbf{v} = 0$ (Homogeneous Linear System), to have a non-trivial $\mathbf{v}$

$$\det(A - \lambda I) = 0$$

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For an eigenvalue problem $(A - \lambda I)\mathbf{v} = 0$ (Homogeneous Linear System), to have a **non-trivial** $\mathbf{v} \neq 0$

$$\det(A - \lambda I) = 0$$

Steps to find eigenvalue λ and eigenvector $\mathbf{v}$ of Matrix A

STEP-1 Find the **characteristic polynomial**

$$\mathbf{P}(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) \text{ of } A$$

STEP-2 Find the roots of $\mathbf{P}(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = \mathbf{0}$ to obtain the **eigenvalues** λ of A .

STEP-3 Use **eigenvalues** λ (one-by-one) in $(\mathbf{A} - \lambda\mathbf{I}) \mathbf{v} = \mathbf{0}$ to find **eigenvectors** $\mathbf{v}$ of A , as follows

- ① Form the matrix M by subtracting λ down the diagonal of A .
- ② Find a basis for the solution space of the homogeneous system $M\mathbf{v} = \mathbf{0}$.

① Find λ from $\det(\mathbf{A} - \lambda\mathbf{I}) = \mathbf{0}$

② Find $\mathbf{v}$ by $(\mathbf{A} - \lambda\mathbf{I}) \mathbf{v} = \mathbf{0}$, for each λ

Characteristic Polynomial $P(\lambda) = \det(A - \lambda I)$

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then $(A - \lambda I) = \begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{bmatrix}$ and

Characteristic Polynomial is given by

$$\begin{aligned} P(\lambda) &= \det(A - \lambda I) = \det \begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{bmatrix} \\ &= (a_{11} - \lambda)(a_{22} - \lambda) - (a_{12})(a_{21}) \\ &= \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{12}a_{21}) \\ &= \lambda^2 - (\text{sum of diagonal entries of } A)\lambda + (\text{determinant of } A) \end{aligned}$$

$$P(\lambda) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$$

Find eigenvalues of $A = \begin{bmatrix} 3 & -4 \\ 2 & -6 \end{bmatrix}$

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$$P(\lambda) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$$

We know that $\text{tr}(A) = 3 + (-6) = -3$ and $\det(A) = -10$

$$P(\lambda) = \lambda^2 + 3\lambda - 10$$

Characteristic Equation is

$$P(\lambda) = 0 \quad \text{or} \quad \lambda^2 + 3\lambda - 10 = 0 \implies \lambda =$$

Eigenvectors ?

For given λ we can find its corresponding eigenvectors by putting the value of λ in

$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$

Thus for $A = \begin{bmatrix} 3 & -4 \\ 2 & -6 \end{bmatrix}$, we have $\lambda = 2, -5$

(For $\lambda = 2$): we have to solve $(A - 2I)\mathbf{v} = \mathbf{0}$

(For $\lambda = -5$): we have to solve $(A - (-5)I)\mathbf{v} = \mathbf{0}$

[(For $\lambda = 2$):] we have to solve $(A - 2I)\mathbf{v} = \mathbf{0}$

Here $A = \begin{bmatrix} 3 & -4 \\ 2 & -6 \end{bmatrix}$ this means that

$$(A - 2I)\mathbf{v} = \mathbf{0} \implies \begin{bmatrix} 3-2 & -4 \\ 2 & -6-2 \end{bmatrix} \mathbf{v} = \mathbf{0}$$

or

$$\begin{bmatrix} 1 & -4 \\ 2 & -8 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Row reduction algorithm implies that

$$\begin{bmatrix} 1 & -4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \implies v_1 - 4v_2 = 0$$

This system has infinity many solution of the form of $\begin{bmatrix} 4t \\ t \end{bmatrix}$ $t \in \mathbf{R}$.

Suppose $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$. Then

$$P(t) = t^3 - \text{tr}(A) t^2 + (A_{11} + A_{22} + A_{33})t - \det(A)$$

(Here A_{11} , A_{22} , A_{33} denote, respectively, the cofactors of a_{11} , a_{22} , a_{33} .)

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Find the characteristic polynomial of $A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 3 & 2 \\ 1 & 3 & 9 \end{bmatrix}$.

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$$P(t) = t^3 - \text{tr}(A) t^2 + (A_{11} + A_{22} + A_{33})t - \det(A)$$

We have $\text{tr}(A) = 1 + 3 + 9 = 13$.

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$$A_{11} = \begin{vmatrix} 3 & 2 \\ 3 & 9 \end{vmatrix} = 21, \quad A_{22} = \begin{vmatrix} 1 & 2 \\ 1 & 9 \end{vmatrix} = 7, \quad A_{33} = \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} = 3$$

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Thus, $A_{11} + A_{22} + A_{33} = 31$. Also, $|A| = 27 + 2 + 0 - 6 - 6 - 0 = 17$. Accordingly,

Suppose $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$. Then

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Thus, $A_{11} + A_{22} + A_{33} = 31$. Also, $|A| = 27 + 2 + 0 - 6 - 6 - 0 = 17$. Accordingly,

$$P(t) = t^3 - 13t^2 + 31t - 17$$

Properties of Eigenvalues of A

$$\text{tr}(A) = \sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i$$

$$\det(A) = \prod_{i=1}^n \lambda_i$$

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -4 \\ 2 & -6 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}.$$

Verify that

- ① $\det(A) = \text{Product of eigenvalues of } A$
- ② $\det(B) = \text{Product of eigenvalues of } B$
- ③ $\det(C) = \text{Product of eigenvalues of } C$
- ④ $\text{tr}(A) = \text{Sum of eigenvalues of } A$
- ⑤ $\text{tr}(B) = \text{Sum of eigenvalues of } B$
- ⑥ $\text{tr}(C) = \text{Sum of eigenvalues of } C$

Every Matrix Transformations is Linear

Example

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the "**projection**" mapping into the xy -plane; that is, T is the mapping defined by

$$T(x, y, z) = (x, y, 0).$$

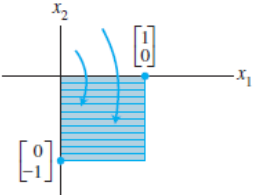
We have shown that T is linear.

If $T = A$, where A is given by

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{then} \quad T\mathbf{x} = A\mathbf{x} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$$

Calculate the eigenvalues and eigenvectors of **Projection** matrix A

$$\det(A - \lambda I) = \begin{vmatrix} 1 - \lambda & 0 & 0 \\ 0 & 1 - \lambda & 0 \\ 0 & 0 & -\lambda \end{vmatrix} = 0 \implies \lambda(\lambda - 1) = \boxed{\lambda = 0, 1}$$

Transformation	Image of the Unit Square	Standard Matrix
Reflection through the x_1 -axis		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$$\det(A - \lambda I) = \begin{vmatrix} 1 - \lambda & 0 \\ 0 & -1 - \lambda \end{vmatrix} = 0 \implies (\lambda + 1)(\lambda - 1) = \boxed{\lambda = -1, 1}$$

$$A = [T] = \begin{bmatrix} 2 & 1 & -2 \\ 2 & 3 & -4 \\ 1 & 1 & -1 \end{bmatrix} \quad \text{and} \quad \begin{aligned} \text{tr}(A) &= 4, & |A| &= 2 \\ A_{11} &= 1, & A_{22} &= 0, & A_{33} &= 4 \\ \sum_i A_{ii} &= 5 \end{aligned}$$

Therefore, $\Delta(t) = t^3 - 4t^2 + 5t - 2 = (t - 1)^2(t - 2)$, and so $\lambda = 1$ and $\lambda = 2$ are the eigenvalues of A (and T). We next find linearly independent eigenvectors for each eigenvalue of A .

- (i) Subtract $\lambda = 1$ down the diagonal of A to obtain the matrix

$$M = \begin{bmatrix} 1 & 1 & -2 \\ 2 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}, \quad \text{corresponding to} \quad x + y - 2z = 0$$

Here y and z are free variables, and so there are two linearly independent eigenvectors belonging to $\lambda = 1$. For example, $u = (1, -1, 0)$ and $v = (2, 0, 1)$ are two such eigenvectors.

- (ii) Subtract $\lambda = 2$ down the diagonal of A to obtain

$$M = \begin{bmatrix} 0 & 1 & -2 \\ 2 & 1 & -4 \\ 1 & 1 & -3 \end{bmatrix}, \quad \text{corresponding to} \quad \begin{aligned} y - 2z &= 0 \\ 2x + y - 4z &= 0 \\ x + y - 3z &= 0 \end{aligned} \quad \text{or} \quad \begin{aligned} x + y - 3z &= 0 \\ y - 2z &= 0 \end{aligned}$$

Only z is a free variable. Here $w = (1, 2, 1)$ is a solution.

Thus, T is diagonalizable, because it has three independent eigenvectors. Specifically, choosing

$$S = \{u, v, w\} = \{(1, -1, 0), (2, 0, 1), (1, 2, 1)\}$$

$$AA^T = \begin{bmatrix} 5 & -2 & 1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad A^T A = \begin{bmatrix} 2 & -2 \\ -2 & 5 \end{bmatrix}, \quad BB^T = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix},$$

$$\& \quad B^T B = \begin{bmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{bmatrix}$$

Match the Eigenvectors to the corresponding Matrix and find Eigenvalues for each case

① $\begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

② $\begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

③ $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix}$

④ $\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$

Eigenvectors and Eigenvalues of A

Let

$$A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}, \quad v_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then:

$$Av_1 = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} = v_1$$

$$Av_2 = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4v_2$$

Thus:

v_1 and v_2 are eigenvectors with $\lambda_1 = 1$ and $\lambda_2 = 4$.

Similar Matrices

If A and B are $n \times n$ matrices, then A is similar to B if there is an invertible matrix P such that

$$P^{-1}AP = B, \text{ or, equivalently, } A = PBP^{-1}.$$

If $n \times n$ matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

Warning

Similarity is not the same as row equivalence. (If A is row equivalent to B , then $B = EA$ for some invertible matrix E .) Row operations on a matrix usually change its eigenvalues

$$A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}, \quad v_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Let:

$$P = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

For $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is:

$$P^{-1} = \frac{1}{\det(P)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

where $\det(P) = ad - bc$.

Then:

$$B = P^{-1}AP = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

Factorization and Powers of A

The factorization of A is:

$$A = PDP^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Using this factorization, the fourth power of A is:

$$A^4 = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 256 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 171 & 85 \\ 170 & 86 \end{bmatrix}$$

Polynomial Function of A

For $f(t) = t^3 - 5t^2 + 3t + 6$, we have $f(1) = 5$ and $f(4) = 2$. Then:

$$f(A) = Pf(D)P^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Simplifying:

$$f(A) = \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}$$

Square Root of A

The positive square root of A is:

$$B = P\sqrt{D}P^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Simplifying:

$$B = \begin{bmatrix} \frac{5}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{4}{3} \end{bmatrix}, \quad \text{where } B^2 = A$$

SVD

Singular Value
Decomposition

- If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, then we regard them as $n \times 1$ column matrices.
- The transpose $\mathbf{u}^T$ is a $1 \times n$ row matrix,
- The matrix product $\mathbf{u}^T \mathbf{v}$ is a 1×1 matrix, which we write as a scalar.

Inner Product

The inner product of $\mathbf{u}$ and $\mathbf{v}$, often written as $\mathbf{u} \cdot \mathbf{v}$, is also referred to as the **dot product**.

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

The inner product is:

$$\begin{aligned} \mathbf{u}^T \mathbf{v} &= \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \\ &= u_1 v_1 + u_2 v_2 + \cdots + u_n v_n \end{aligned}$$

Example: Inner Product for Matrices

Consider the matrices A and B :

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

The inner product is defined as:

$$A^T B$$

First, compute the transpose of A :

$$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Next, compute $A^T B$:

$$A^T B = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 26 & 30 \\ 38 & 44 \end{bmatrix}$$

An $m \times n$ matrix U has orthonormal columns if and only if

$$U^T U = I$$

.

Does $U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & -\frac{2}{3} \\ 0 & \frac{1}{3} \end{bmatrix}$ has orthonormal columns?

$$U^T U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{3} \\ \frac{1}{\sqrt{2}} & -\frac{2}{3} \\ 0 & \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Definition: An **orthogonal matrix** is a square matrix Q with real entries that satisfies:

$$Q^T Q = Q Q^T = I$$

Where:

- Q^T : Transpose of Q ,
- I : Identity matrix.

Properties:

- 1 The inverse of Q is its transpose: $Q^{-1} = Q^T$.
- 2 Determinant: $\det(Q) = \pm 1$.
- 3 Preserves Euclidean norm: $\|Qv\| = \|v\|$ for any vector v .

Example of Orthogonal Matrix

Matrix:

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Check Orthogonality:

- Compute Q^T :

$$Q^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

- Compute $Q^T Q$:

$$Q^T Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Since $Q^T Q = I$, Q is an orthogonal matrix.

Applications of Orthogonal Matrices

- Preserve vector lengths during transformations.
- Used in QR decomposition in numerical analysis.
- Represent rotations and reflections in geometry.

Theorem (SVD Theorem).

Let $A \in \mathbb{R}^{m \times n}$ be a rectangular matrix of rank $r \in [0, \min(m, n)]$.
The SVD of A is a decomposition of the form

$$\begin{matrix} n \\ \boxed{A} \\ m \end{matrix} = \begin{matrix} m \\ \boxed{U} \\ m \end{matrix} \begin{matrix} n \\ \boxed{\Sigma} \\ m \end{matrix} \begin{matrix} n \\ \boxed{V^T} \\ n \end{matrix}$$

with an orthogonal matrix $U \in \mathbb{R}^{m \times m}$ with column vectors $u_i, i = 1, \dots, m$,
and an orthogonal matrix $V \in \mathbb{R}^{n \times n}$ with column vectors $v_j, j = 1, \dots, n$.
Moreover, Σ is an $m \times n$ matrix with $\Sigma_{ii} = \sigma_i \geq 0$ and $\Sigma_{ij} = 0, i \neq j$.

Singular Values are the diagonal entries $\sigma_i, i = 1, \dots, r$, of Σ .

Left-Singular Vectors u_i are called the Left-Singular Vectors

Right-Singular Vectors v_j are called the Right-Singular Vectors

$$\begin{aligned}
 \mathbf{A} &= \begin{bmatrix} 1 & -0.8 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \\
 &= \begin{bmatrix} -0.79 & 0 & -0.62 \\ 0.38 & -0.78 & -0.49 \\ -0.48 & -0.62 & 0.62 \end{bmatrix} \begin{bmatrix} 1.62 & 0 \\ 0 & 1.0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -0.78 & 0.62 \\ -0.62 & -0.78 \end{bmatrix}
 \end{aligned}$$

The SVD of a matrix A can be written as:

$$A = U\Sigma V^T$$

Where:

- U is an orthogonal matrix whose columns are the left singular vectors of A ,
- Σ is a diagonal matrix containing the singular values of A ,
- V is an orthogonal matrix whose columns are the right singular vectors of A .

Steps for finding the SVD:

- 1 Compute $A^T A$ to find the right singular vectors V .
- 2 Compute AA^T to find the left singular vectors U .
- 3 Compute the singular values from the eigenvalues of $A^T A$ (or AA^T).

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 5 & -2 & 1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad \& \quad A^T A = \begin{bmatrix} 2 & -2 \\ -2 & 5 \end{bmatrix}$$

① $\begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ are Eigenvectors of AA^T corresponding Eigenvalues, 6, 1 and 0.

② $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ are Eigenvectors of $A^T A$ corresponding Eigenvalues, 6 and 0.

